

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g. Star, Bus, Mesh)
- Clearly label all components such as Router, Switch, Access point, and end devices (PCs, computer laptops etc)

⇒ On the basis of physical layout and structural inspection of college campus network at our organisation, the network uses a combination of Star topology and Extended tree topology.

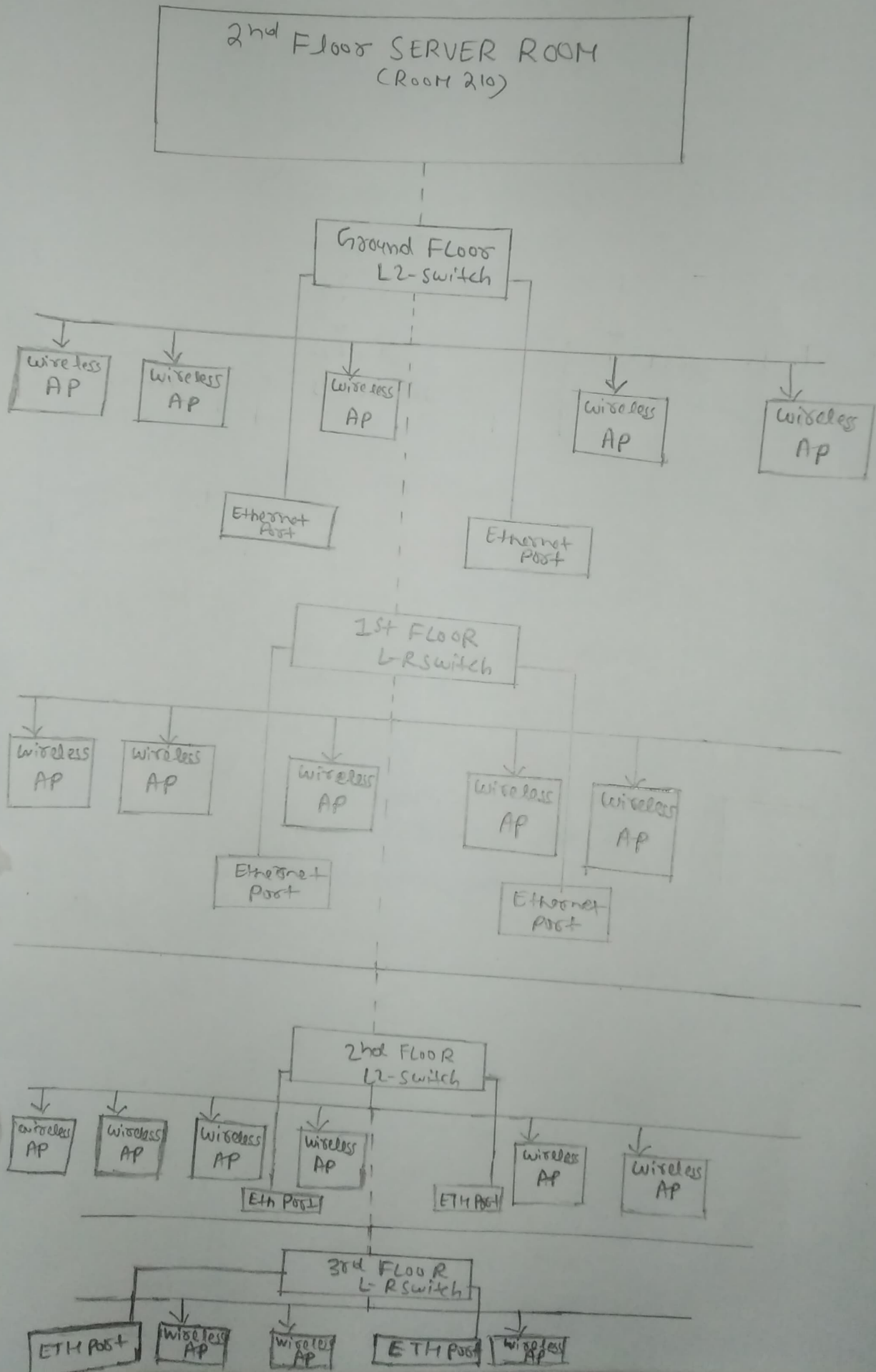
1. Central switch (Star topology)

- A single switch acts as the central networking device
- All three floors are directly connected to the switch at each floor, also ground floor
- Since, every floor connects to a central device, it forms a star topology

2. Floor-wise Distribution (Tree topology)

- The network is divided into three floors
- Each floor contains five routers or three routers
- These routers further connect to computers, laptops, printers, or other devices
- The hierarchical arrangement of switch → Floor → Routers → End devices forms an extended tree topology

HIERARCHICAL STAR TOPOLOGY



2. Identify Network Devices:

- List all Networking devices used in LAN
- Mention the function of each devices

Our campus uses different networking devices a mix of routing distribution, and access layer devices to maintain Seamless connectivity. Below is the list of all networking devices identified in the LAN with their Specific functions:-

- Central switch: Acts as the central backbone device. It handles internal data switching between the routers using physical MAC addresses, ensuring fast packet delivery within the main campus/ building network.

Function → • It acts as the central connection point or point for a specific floor, connecting local end devices like PCs and wifi Access points to the network

- It supports L2+ Management and VLANs (virtual LANs) to segment network traffic to prevent broadcast storms.
- It has 4-port 10GbE SFP+ slots where the yellow fibre optic uplink cables are connected to deliver high-speed backbone bandwidth from the server room

- Routers: Separate the main network into distinct subnets or departments. They manage traffic routing, assign local IP addresses, and provide security and firewall boundaries for the devices connected under them

Function → • It acts as the primary gateway between the internal college LAN and the external WAN (Wide Area Network) or Internet Service provider

- It handles routing decisions by routing data packets into and out of the campus network
- It manages (see Network Security, Network Access Translation), and traffic policies

- Access Point (AP): Same devices as router in college

Function → A wireless access point operates at Layer 2 (Data link layer) and acts as a bridge between the wired Ethernet network and wireless wifi devices.

Its specific function are:

- Receives the wired Ethernet signal from the switch and converts it into a wireless wifi signal using radio waves (2.4 GHz or 5 GHz frequency)
- Manages multiple wireless connections simultaneously using SSID (Service Set Identifier) - the wifi network name
- Controls wireless security using WPA2 encryption to prevent unauthorized access

- Wired End Device (PC): A desktop PC is an end device that operates at Layer 7 (Application Layer). It is the device that actually generates and consumes data on the Network.

Function →

- Sends and receives data over the network using a NIC (Network Interface Card) via Ethernet cable.
- Has a unique MAC address (hardware address) used by the switch to identify it.
- It is assigned an IP address (either static or via DHCP from the router) used for communication across the network.
- Connects directly to the Floor Switch via Cat 6 Ethernet cable, bypassing the router for local traffic.

- Wireless End Device (Laptop / Mobile phone): A laptop or mobile is also an end device but connects wirelessly instead of via cable.

Function →

- Connects to the nearest router/AP using WiFi over radio waves.
- Sends requests to the router which forwards them to the entire internet or other LAN devices.
- Has both a MAC address (wireless NIC hardware address) and an IP address assigned by DHCP.

Role of Subnet mask in network

- Ethernet cable (Cat6): Cat6 (Category 6) cable is the physical transmission medium used throughout the wired backbone of the college LAN

- Function →
- Carries electrical signals encoding data between devices at speeds up to 1 Gbps (Gigabit Ethernet)
 - Connects the ISP router to the building backbone, backbone to floor switches, switches to routers, and switches to desktop PCs
 - Has 8 copper wires twisted in 4 pairs inside - the twisting reduces electromagnetic interference (EMI) between cables
 - Uses RJ-45 connectors at both ends to plug device ports

3. Analyse connection types:

- Specify whether connections are
 - wired (Ethernet)
 - wireless (wifi)
- Mention where each type is used

In our college's LAN there are two types of connections used at different levels of the network. Every connection in the entire building is either wired (Ethernet) or wireless (WiFi). Both types work together to form the complete network — wired forms backbone and distribution layer, wireless forms the access layer where users actually connect.

Wired Connection (Ethernet): It uses a physical Cat 6 Ethernet cable to transmit data as electrical signals between two devices. It follows the IEEE 802.3 standard. It has 8 copper wire twisted in 4 pairs inside, with RJ-45 connectors at both ends that plug into device ports.

How it works: When a PC sends data, it converts the data into electrical signals and pushes them through the copper wire of the Cat 6 cable. The switch at the other end receives these signals, reads the destination signals — MAC address, and forwards the data to the correct port. This entire process happens in milliseconds.

Where it is used in our college LAN

<u>Connection point</u>	<u>Description</u>
ISP → Main Router	WAN Link bringing internet into the building
Main Router → Building backbone	Vertical trunk connecting all 4 floors
Backbone → Floor Switch	One tap per floor from the backbone
Floor Switch → Router/AP per floor	Star connection on each floor
Floor Switch → Desktop PC	Direct Ethernet ports for wired desktops

Wireless connection (WiFi): It uses radio waves to transmit data through the air between a Router/AP and end devices like laptops and mobile phones. No physical cable is needed between AP and the end device.

How it works: The router/AP receives data from the switch via Ethernet cable, converts it into radio wave signals, and broadcasts them in all directions within its coverage zone. A laptop's wireless NIC picks up these radio waves, converts them back into data, and passes them to the operating system. The frequency used is either 2.4 GHz or 5 Hz.

where it is used in our college LAN

Connection point

Router/AP $\rightarrow$ Laptops

Router/AP $\rightarrow$ Mobile phones

Router/AP $\rightarrow$ Any wifi device

Description

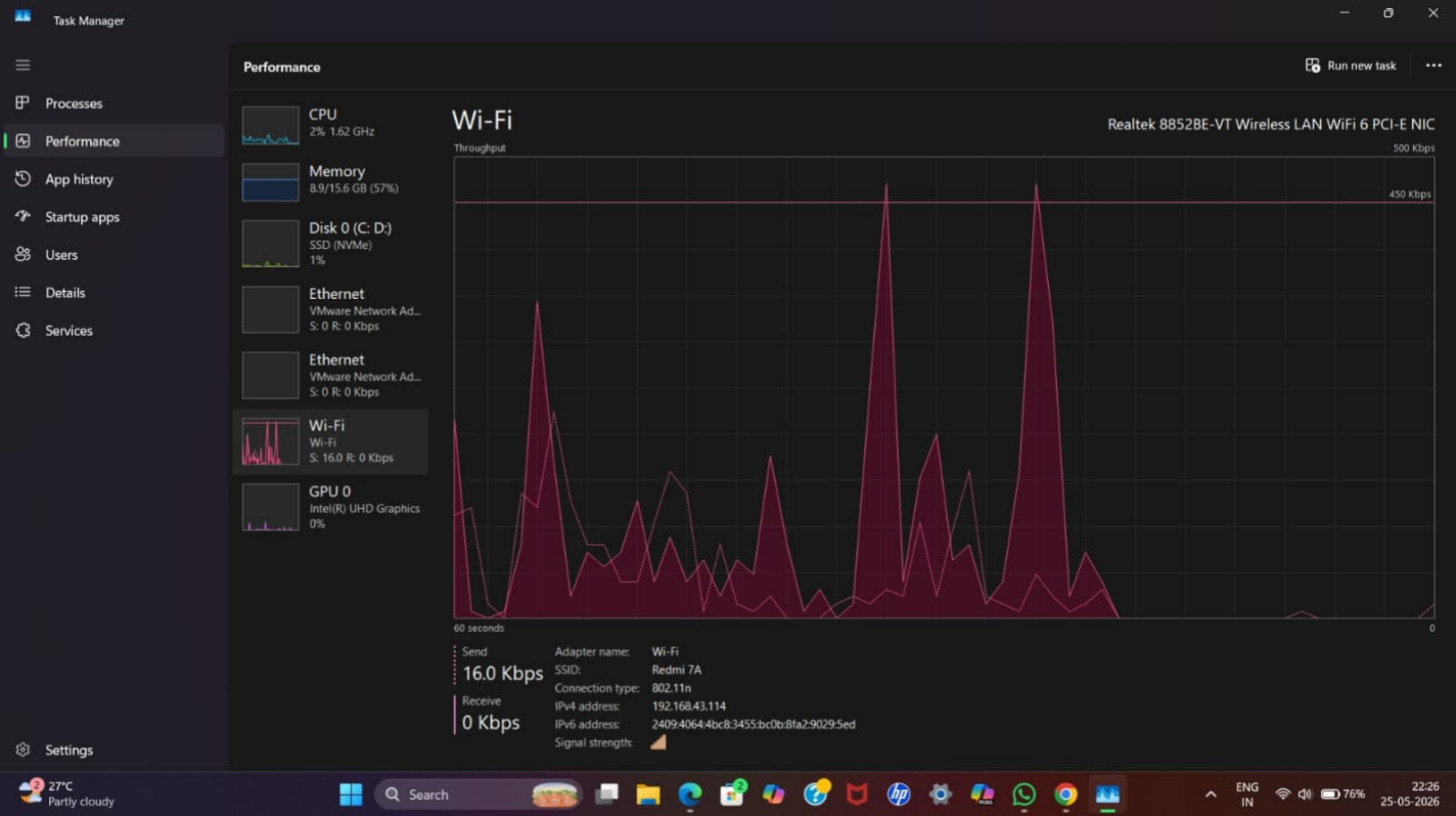
Student's laptops connect wirelessly in classrooms and labs

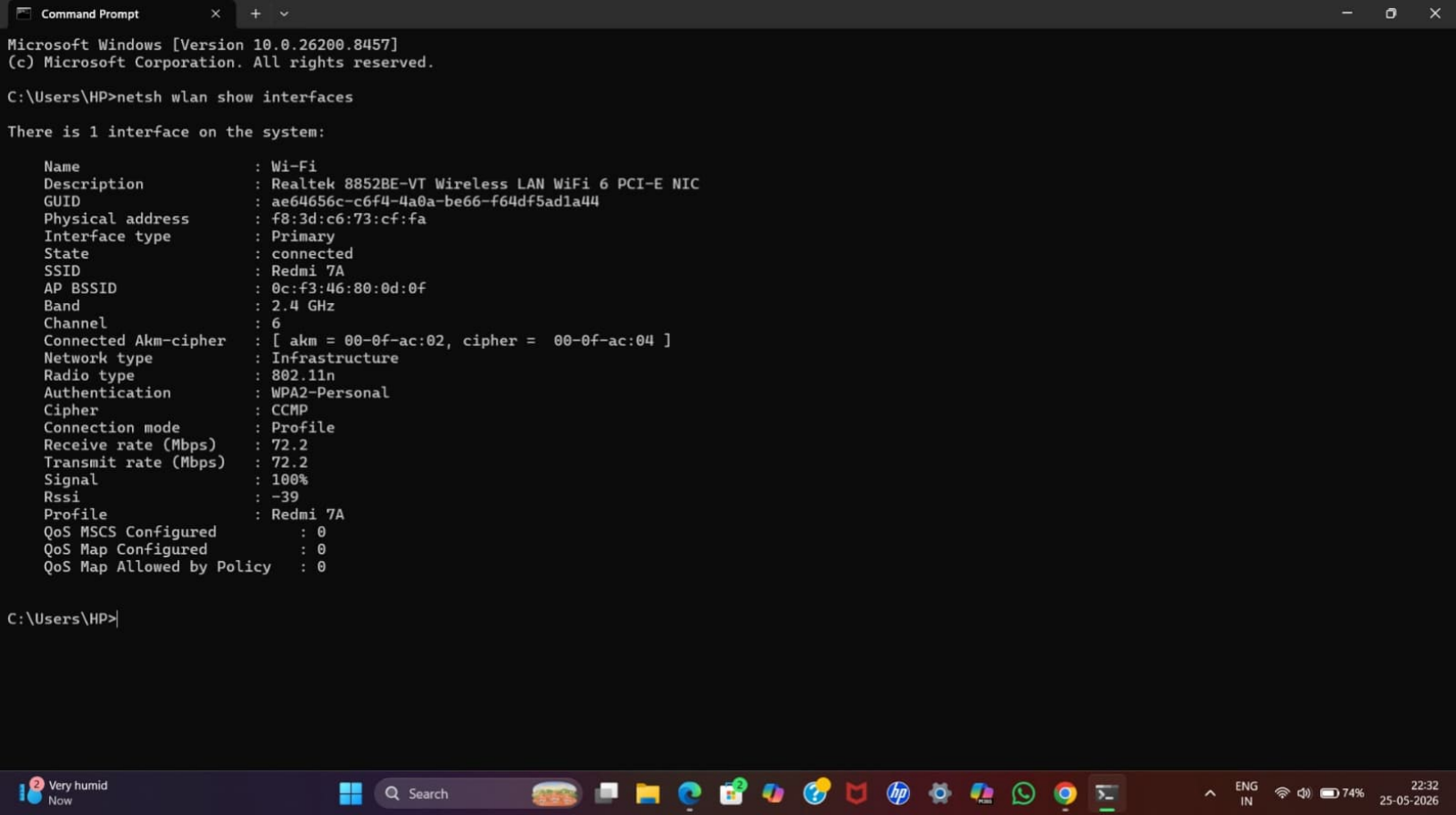
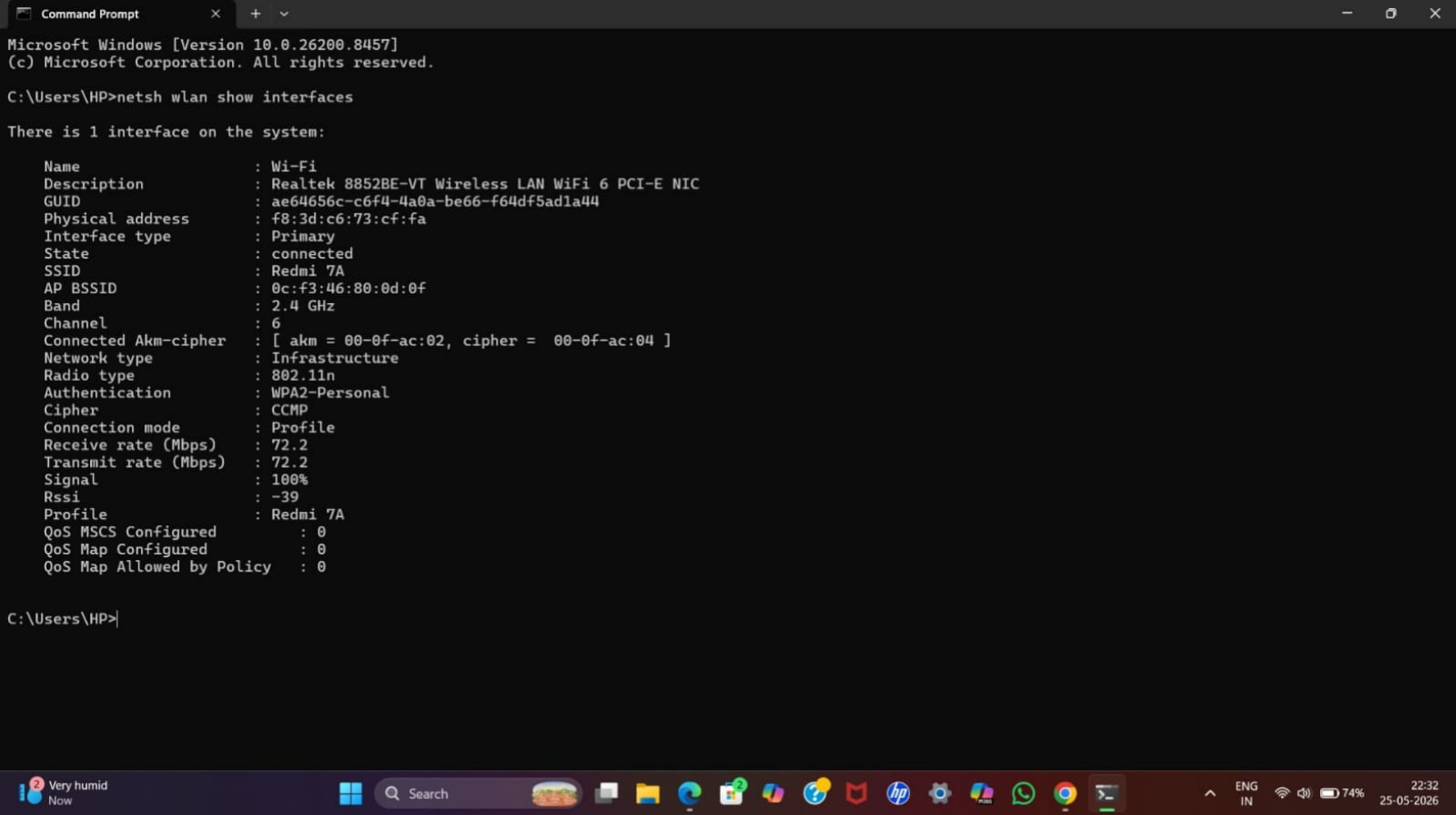
Staff and Student mobiles connect to college wifi

Any device with a wireless NIC within the AP's range

4. Measure Network Speed:

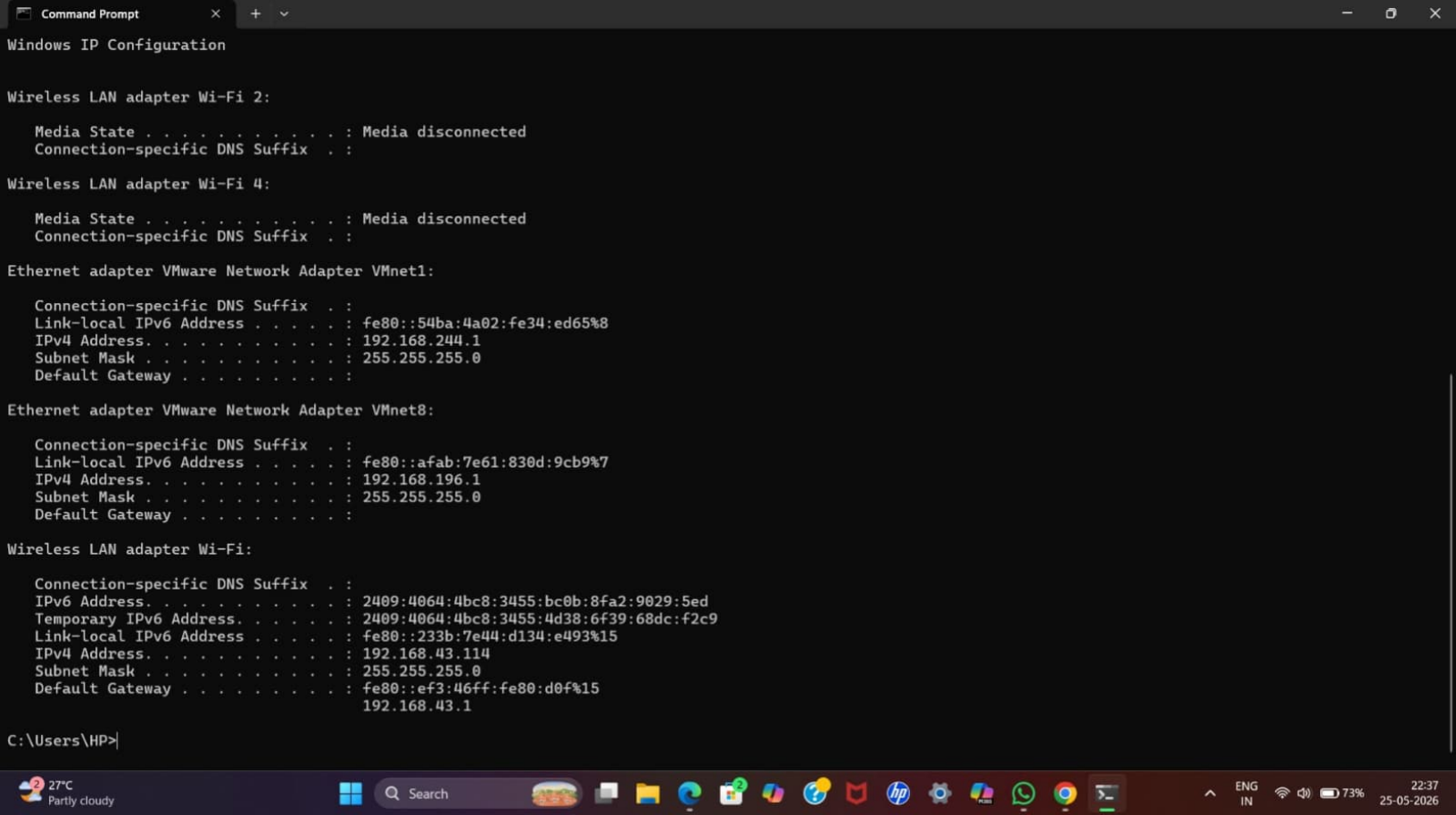
- Find the link speed of:
 - wired network (if available)
 - wireless network
- Mention the tools / commands used (e.g. Task Manager, netsh; netsh wlan show interfaces);

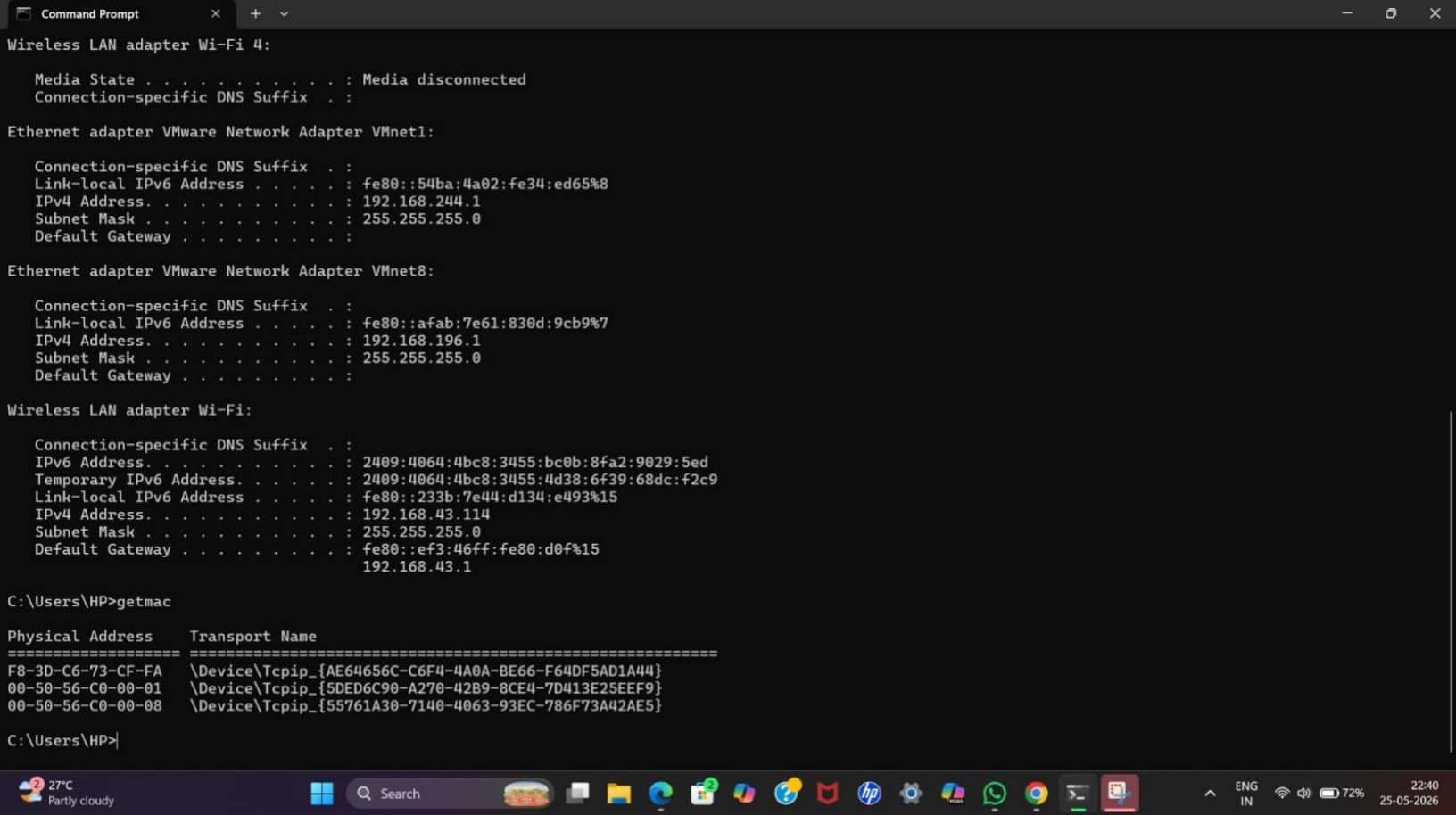




5. Find System Network Details:

- Determine and record the following of your computer
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
- Mention the command(s) used (e.g., `ipconfig`, `getmac`).





```
Command Prompt

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::54ba:4a02:fe34:ed65%8
IPv4 Address. . . . . : 192.168.244.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::afab:7e61:830d:9cb9%7
IPv4 Address. . . . . : 192.168.196.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:4064:4bc8:3455:bc0b:8fa2:9029:5ed
Temporary IPv6 Address. . . . . : 2409:4064:4bc8:3455:4d38:6f39:68dc:f2c9
Link-local IPv6 Address . . . . . : fe80::233b:7e44:d134:e493%15
IPv4 Address. . . . . : 192.168.43.114
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::ef3:46ff:fe80:d0f%15
                             192.168.43.1

C:\Users\HP>getmac

Physical Address      Transport Name
=====
F8-3D-C6-73-CF-FA    \Device\Tcpip_{AE64656C-C6F4-4A0A-BE66-F64DF5AD1A44}
00-50-56-C0-00-01    \Device\Tcpip_{5DED6C90-A270-42B9-8CE4-7D413E25EEF9}
00-50-56-C0-00-08    \Device\Tcpip_{55761A30-7140-4063-93EC-786F73A42AE5}

C:\Users\HP>
```


6. Explain Observations

• Briefly explain;

- Difference between wired and wireless speed
- Role of subnet mask in networking

Wired Network

- Faster Communication
- Stable Connection
- Low latency
- Dedicated physical medium
- Actual real-world Speed 900-950 Mbps
- Maximum theoretical Speed 1 Gbps (Cat6)
- Standard: IEEE 802.3

Wireless Network

- Slower Communication
- Affected by distance/Interference
- Higher latency
- Shared radio medium
- 50-500 Mbps
- Maximum theoretical Speed 300 Mbps - 1.3 Gbps
- Standard: 802.11n/802.11ac

- Ethernet provide physical communication using network cables resulting in higher speed & stable performance
- wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users

Role of Subnet mask in network

A Subnet mask is a 32-bit number used with an IP address to divide it into two parts:

- the network ID and
- the host ID

In a Subnet mask:

- Binary 1s represent the network portion of the IP address
- Binary 0s represent the host portion

It helps a device determine whether another device is on the same local network or on a different network, which in turn decides whether communication is direct or must go through a router.

Function of Subnet Mask:

- Separates an IP address into network ID and host ID
- Divides a large network into smaller subnets for better organization
- Improves network efficiency, security, and manageability by reducing unnecessary traffic
- Helps routers determine whether a packet should be forwarded locally or to another network